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Le Chatelier's Principal Lab Report

Purpose:

The purpose of this experiment was to investigate the influence of temperature and different reagents on chemical equilibria. Specifically, the equilibria of $[CuBr_4]^{2-}$, $Fe(SCN)^{2+}$, and the ionization of a weak acid, CH_3COOH, were examined using Le Chatelier's principle. Le Chatelier's principle helps predict and explain how equilibrium systems respond to specific changes.

Procedure:

The $[CuBr_4]^{2-}$ Equilibrium System

A 50 mL beaker was filled with 15 mL of 0.20M $CuSO_4$ solution, and the initial color was recorded.

5 mL of saturated KBr solution was added to the same beaker, and the resulting color change was noted.

The solution was divided equally into three test tubes. One tube was kept at room temperature, and its color was observed.

Another tube was placed in a hot water bath until a color change occurred, and the new color was recorded.

The third tube was placed in an ice-water bath until a color change occurred, and the observations were recorded.

The $Fe(SCN)^{2+}$ Equilibrium System

50 mL of DI water was placed in a 100 mL beaker, and 20 drops of $Fe(NO_3)$ and 20 drops of NH_4SCN were added, resulting in a uniform color that was noted.

Four test tubes were partially filled with the equilibrium mixture from step one. One tube served as a control.

The second tube had 10 drops of Fe(NO3)3 added, the third tube had 10 drops of NH4SCN added, and the fourth tube had 10 drops of oxalic acid added. The colors of each solution were compared to the control, and observations were recorded.

Ionization of a Weak Base

15 mL of deionized water were added to a 100 mL beaker, followed by two drops of 15M NH_3 and two drops of phenolphthalein indicator. The solution was thoroughly mixed to ensure uniformity. Then, divided equally into two separate test tubes. Retain one test tube as a color control for comparison.

In the second test tube, 15 drops of 2M NH4Cl solution were added. Gently shake the test tube to thoroughly mix the contents.

Ionization of a Weak Acid

Two drops of CH_3COOH and two drops of methyl orange indicator were mixed with 15 mL of deionized water in a 100 mL beaker.

The solution was divided into two test tubes. One tube served as a control.

Five drops of $NaCH_3COO$ solution were added to the second test tube, and the observations were recorded.

Results and Observations:

Part 1: The $[CuBr_4]^{2-}$ Equilibrium System

$$[Cu(H20)_6]^{2+}(aq) + 4\ Br^-(aq) \leftrightarrow [CuBr_4]^{2-}(aq) + 6\ H20(l)$$

Blue *yellow*

Control Solution or Change Imposed	Observations	Explanations
$CuSO_4(aq) + KBr(aq)$	The solution is turquoise	The addition of KBr resulted in a turquoise color, indicating a shift to the right of the equilibrium.
Room temperature	The solution remained turquoise	At room temperature, the solution remained turquoise, suggesting equilibrium.
Hot water bath	The solution turned an olive-green color.	In the hot water bath, the solution turned olive-green, indicating an endothermic reaction and a shift to the left.
Coldest		In the ice-water bath, the solution became lighter blue, suggesting an exothermic reaction and a shift to the right.

Part 2: The $Fe(SCN)^{2+}$ Equilibrium System

$$Fe^{3+}(aq) + SCN^{-}(aq) \leftrightarrow [FeSCN]^{2+}(aq)$$

Control Solution or Change Imposed	Observations	Explanations
NH_4SCN + $Fe(NO_3)_3$	Dark Orange/Brown	The addition of Fe(NO3)3 and NH4SCN resulted in a yellow, dark orange/brown color, indicating a shift to the right of the equilibrium.
0.1M $Fe(NO_3)_3$	Burnt Orange	In the second test tube with added Fe(NO3)3, the solution turned burnt orange, consuming the added Fe3+.
0.1M NH_4SCN	Dark Red	In the third test tube with added NH4SCN, the solution turned dark red, consuming the added SCN-.
0.1M $H_2C_2O_4(aq)$	Clear @ bottom Yellow @ middle Orange @ top	In the fourth test tube with added oxalic acid, the solution exhibited a clear bottom, yellow in the middle, and orange at the top, indicating a shift to the left and an increase in Fe3+.

Part III: Ionization of a Weak Base

$$NH_3(aq) + H_20(l) \leftrightarrow NH_4^{+}(aq) + 0H^{-}(aq)$$

Control Solution or Change Imposed	Observations	Explanations
NH_3 (control) + phenolphthalein	Very light pink	
NH_4Cl	Clear	The addition of NH_4Cl resulted in the solution turning clear, indicating a shift to the left of the equilibrium

Part 4: Ionization of a Weak Acid

$$CH_3C00H(aq) + H_20(l) \leftrightarrow H_30^{+}(aq) + CH_3C00^{-}(aq)$$

Control Solution or Change Imposed	Observations	Explanations
CH_3COOH + methyl orange indicator	Slight pink	
$NaCH_3COO$	Saturated yellow	When sodium acetate is added to a solution of acetic acid, the equilibrium shifts to the left. The color of the solution turns yellow due to the formation of acetate ion from sodium acetate. The dissociation of acetic acid reduces and the pH increases.